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MILITARY SPECIFICATION

PROPELLANTS, NITROGEN TETROXIDE

This specification is mandatory for use by all departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Purpose This specification covers the requirements for seven types of nitrogen tetroxide (N_2O_4) propellants

1.2 Classification Nitrogen tetroxide propellants shall be of the following designated compositions as specified (6.2).

Compositions

NTO - Nominal 99.5% N_2O_4 purity with red-brown color

MON-1 - Nominal 99% N_2O_4 and 1% NO with green color.

Normal Grade - no iron requirement.
Low-Iron Grade - maximum 0.5 parts per million iron

MON-3 - Nominal 97% N_2O_4 and 3% NO with green color.

Normal Grade - no iron requirement
Low-Iron Grade - maximum 0.5 parts per million iron

MON-10 - Nominal 90% N_2O_4 and 10% NO with green color.

MON-25 - Nominal 75% N_2O_4 and 25% NO with green color

2.0 APPLICABLE DOCUMENTS

2.1 Defense Standardization Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Code (68) SA-ALC/SFRM, Kelly AFB TX 78241-5000, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9135

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SPECIFICATIONS

Military

MIL-P-25508	Propellant, Oxygen
MIL-P-27401	Propellant, Pressurizing Agent, Nitrogen
MIL-P-27407	Propellant, Pressurizing Agent, Helium
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene with Dispenser

STANDARDS

Military

MIL-STD-101	Color Code for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other Publications The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

Department of Transportation (DOT)

40 CFR 170	Administrative procedures of the Department of Transportation
49 CFR 171-190	Hazardous Materials Regulations of the Department of Transportation
DOT P 5800 2	Hazardous Materials Emergency Response Guidebook (1980) Guide No 20 - Nitrogen Tetroxide

American National Standard Institute (ANSI)

ANSI Z129 1	Precautionary Labeling of Hazardous Industrial Chemicals (1976).

(Application for copies should be addressed to the American National Standards Institute, New York, New York 10018.)

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American Society for Testing and Material (ASTM).

ASTM E 29	Recommended Practices for Indicating which Figures are to be Considered Significant in Specified Limiting Values.
ASTM E 203	Test for Water Using Karl Fischer Reagent.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3.0 REQUIREMENTS

3 1 Chemical Composition and Physical Properties. The chemical composition and physical properties of the propellants shall conform to those listed in Table I when tested in accordance with the applicable test.

3.2 Limiting Values. The following applies to all specified limits of this specification For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of the "Recommended Practices for Indicating which Figures are to be Considered Significant in Specified Limiting Values" (ASTM E 29).

3 3 Filter. A filter with a 10 nominal and 40 absolute micron rating shall be installed between the manufacturer's plant and the container to be filled for delivery

3.4 Qualitative The propellant shall be a clear, single-phased liquid when examined visually by transmitted light in accordance with 4.5 1 The NTO shall be red-brown in color, the MON compositions shall be green in color.

4 0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 2 Classification of Tests. The inspection and testing of the propellants shall be classified as quality conformance tests.

4 3 Test Conditions The test conditions are described under the individual tests to which they apply

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TABLE I. Chemical Composition and Physical Properties

Composition	NTO (Red-Brown)	MON-1 Normal (Green)	MON-1 Low-Iron (Green)	MON-3 Normal (Green)	MON-3 Low-Iron (Green)	MON-10 (Green)	MON-25 (Green)	Test Para.
N_2O_4 assay % wt, MIN	99.5	-	-	-	-	-	-	4 5.3
NO content % wt, MAX		1.0	1 0	3.0	3.0	11.0	26.0	4.5.2
% wt, MIN	1/	0.6	0.6	2 5	2 5	10.0	25.0	
N_2O_4 + NO % wt, MIN	-	99.5	99.5	99 5	99.5	99.5	99.5	4.5 3
Water equiv % wt, MAX	0 17	0.17	0.17	0 17	0.17	0.17	0 17	4.5.4
Chloride % wt, MAX	0.040 2/	0.040	0.040	0.040	0.040	0 040	0.040	4.5.5
Non-Volatile Residue % wt, MAX	-	-	0.005	-	-	-	-	4 5.6
Iron Content ppm wt, MAX	-	-	0 5	-	0 5	-	-	4 5.7
Particulate mg/l, MAX	10 0	10.0	10.0	10.0	10.0	10.0	10 0	4.5.8

1/ The NO content shall be limited to that which does not change the specified red-brown color of the propellant (3 4)

2/ This test need not be performed on NTO propellant if the material was manufactured by the ammonia-oxidation process

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4.4 Quality Conformance Tests. Quality conformance tests shall consist of:

a. Individual tests. 4 4.1

b. Sampling Tests. 4.4.2

4.4.1 Individual Tests. The propellant shall be subjected to the following test as described under 4.5.

Examination of product. 4 5.1

4 4.2 Sampling Tests. The propellant shall be selected in accordance with 4.4.2.1 and subjected (as appropriate to designated composition) to the following tests as described under 4.5:

a. Nitric oxide content 4.5 2

b Nitrogen tetroxide assay 4.5.3

c. Water equivalent 4.5.4

d. Chloride content 4 5.5

e Non-volatile residue 4 5 6

f Iron content 4.5.7

g Particulate 4 5.8

4.4.2 1 Sampling Plan

4 4.2.1 1 Lot A lot shall consist of one of the following:

a The propellant produced in not more than 24 consecutive hours from a continuous process which is used to fill shipping containers directly from the process output. A continuous process shall be the production of product by continuous input of raw materials and output of finished product by one manufacturer in one plant with no change in manufacturing conditions or materials.

b. The propellant from individual runs of a batch process which is used to fill shipping containers directly from the process output A batch process shall be the production of product by runs from single additions of raw materials which are reacted and purified forming the product.

c The propellant from either or both the continuous and batch processes which is held in a single storage tank and subsequently withdrawn to fill shipping containers. The product will be homogeneous at the time of withdrawal and shall not be added to while being withdrawn After each addition to the storage tank, the contents shall constitute a separate lot.

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4 4 2.1.2 Sample. A sample shall consist of not less than 800 milliliters (ml) of propellant. Two 400 ml samples may be taken for the tests specified in 4 4 2 One sample may be used for tests a through f and the other for test g In the case of cylinder sampling, the sample(s) shall be taken from the same cylinder. Unless otherwise specified, quality conformance tests shall be performed on each sample of the propellant taken directly from the shipping containers. When required, samples shall be forwarded to a laboratory designated by the procuring activity for subjection to the quality conformance tests specified herein

4.4.2.1.2.1 Sampler. The sampler shall be capable of accepting a 400 ml or 800 ml sample and constructed of 300 series stainless steel. The top and bottom valves and any outlets or welds shall also be of 300 series stainless steel Samplers shall be precleaned in a manner such as to preclude contamination of propellant samples See also 6 3.b.

4 4 2.1.3 Cylinders The number of cylinders selected for sampling from each lot of Standard Grade material shall be in accordance with Table II The first and last containers to be filled within a given lot shall be sampled Other samples may be selected at random. Each selected cylinder shall be thoroughly agitated immediately before sampling The propellant from each container sampled shall constitute a separate sample

TABLE II. Sampling for Test

Number of Containers in Lot	Number of Containers to be Sampled
1	1
2-8	2
9-27	3
28-64	4

4 4 2 1.4 Portable Tanks, Cargo Tanks, and Tank Cars. A sample shall be taken from each portable tank, cargo tank, or tank car

4 4 2 1.5 Other Containers Unless otherwise specified, other containers of 100 gallons or less water capacity shall be sampled in accordance with 4.4 2.1.3 Containers greater than 100 gallons water capacity shall be sampled in accordance with 4.4.2.1 4

4.4 2 1.6 Method of Taking Sample. Connect a 1/4 inch stainless steel tube approximately 1-1/2 feet in length to the container's (cylinder or tank) dip-tube valve. Fit one end of the tube with a 1 inch to 1-1/4 inch pipe-thread bushing for attaching to the dip-tube valve and fit the other end to a 1/4 inch tee Fit one leg of the tee with a 1/4 inch needle valve for purging and the other leg for connecting to the sampler inlet valve Attach approximately 5 feet of polyethylene tubing to the purging valve to direct the propellant fumes away from the sampling area or purge directly into an evacuated waste tank. Apply thread tape conforming to MIL-T-27730 to all

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thread connections. Wrap the tape under tension starting three threads back from the end and in the direction with the male pipe thread helix toward thread run-out. Wrap once plus an overlap of 1/2 inch at thread run-out end. Connect the evacuated sampler to the tee at an inclined 45° angle with the inlet valve down Open the dip-tube valve on the purging valve. Purge until approximately 2,000 ml of propellant have been removed to clear the dip-tube Then close the purging valve and open the sampler inlet valve. Open the outlet valve and continue sampling until liquid propellant appears. Close the sampler outlet, inlet, and dip-tube valves Open the purging valve to clear the sample line and disconnect the sampler

4.4 3 Rejection. When any sample of the propellant tested in accordance with 4.5 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.5 Test Methods

4.5.1 Examination of Product The sample (4.4.2 1.2) shall be examined with transmitted light to satisfy the qualitative requirements Place 100 ml of sample in a clean, clear glass container (graduated cylinder or large test tube), cool in an ice-bath for 10 minutes, wipe dry, and examine visually by transmitted light from a light bulb or sunlight. The presence of visible sediment in the sample shall be cause for rejection.

4.5 2 Nitric Oxide (NO) Content. The NO content of the MON sample shall be determined by the procedure specified in paragraph 4.5.2 1, or by the alternate procedure described in paragraph 4.5.2.2. In the event of dispute, the procedure specified in paragraph 4.5 2.1 shall be used as the referee method.

4.5.2.1 Nitric Oxide Assay. The NO content of the sample shall be determined by the following method.

4.5 2.1.1 Tared Sample. Evacuate and weigh to the nearest 0.001 gram (g) a sample bomb of approximately 60 ml capacity, fitted with a needle valve Support the sampler in a vertical position on a ring stand in a hood Connect the sample bomb to the sampler and provide for evacuating the connecting line. Evacuate the connecting line, then open the sample bomb and sampler valves Transfer 40 to 60 g of the sample. Close all valves and then evacuate the connecting line. Disconnect the sample bomb and flush the exposed opening of the valve with nitrogen until residual N_2O_4 disappears. Accurately reweigh the sample bomb to the nearest 0.001 g, obtaining the sample weight by difference. If sample weight is below 40 g, another sample shall be taken

4 5.2 1.2 Procedure Attach a rotameter with stainless steel tubing to the reducing valve of a cylinder of oxygen (O_2) Connect a length of 1/16 inch stainless steel tubing to the exit end of the rotameter Open the cylinder valve and purge residual air from the connecting lines and rotameter Immediately attach the sample bomb to the other end of the 1/16 inch tubing and, using the reducing valve, adjust the oxygen pressure to 30 pounds per square inch (psi) Close the cylinder valve Place the sample bomb in an ice bath and maintain the temperature at approximately 32°F (0°C) After the sample has been allowed to cool for 10 minutes permit O_2 to flow into the sample bomb by first opening the cylinder valve and then the bomb valve. Occasionally shake the bomb to hasten absorption and oxidation. Lack of O_2

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flow through the rotameter indicates completion of reaction. Allow the sample bomb to remain in the ice bath for 15 minutes, with valves open after reaction has been completed. Disconnect the sample bomb, remove from the bath, warm to ambient temperature and wipe dry Place on a balance and record the weight to the nearest 0.001 g.

4.5.2.1.3 Calculation. The percent NO shall be calculated by the following formulas·

$$x = B-A$$

$$y = C-B$$

$$\text{NO, percent by weight} = \frac{[y-0\ 003857\ (z - \frac{x+y}{1.49})]\ 187\ 5}{x}$$

where.

x = sample weight in grams

y = O_2 weight in grams.

z = volume of sample bomb in ml. Determined by difference in weight from evacuated and water filled.

A = weight of sample bomb in grams

B = weight of sample bomb plus sample in grams.

C = weight of sample bomb plus sample and O_2 in grams.

4.5 2 1 4 Reagents and Equipment The following reagents and equipment or their equivalent shall apply as test conditions of 4.5.2 1·

a Reagents

(1) Nitrogen, MIL-P-27401, Type 1

(2) Oxygen, MIL-P-25508, Type 1

b Equipment

(1) Valve, needle, 1/8 inch NPT male inlet X 1/4 inch swagelok outlet, stainless steel.

(2) Bomb, sample, 2 inch diameter with 1/8 inch NPT female, 302 stainless steel, 24 gauge.

(3) Stand, ring

(4) Tubing, polyethylene, length as required

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(5) Rotameter, glass-ball, with 0 to 1.0 liter per minute capability.

(6) Tubing, stainless steel, 1/16 inch diameter.

(7) Bath, ice

(8) Balance, analytical, plus or minus 0.1 mg sensitivity, 0.1 mg accuracy.

4.5.2.2. Alternate Procedure.

4.5.2.2.1 Preparation. Clamp the sampler specified in paragraph 4.4.2.1 2.1 to a stand as shown in Figure 1. Attach a 1/16 inch I.D. length of polytetrafluoroethylene tubing to the outlet valve. Place another length of 1/16 inch I.D. polytetrafluoroethylene tubing in a beaker containing a strong NaOH solution. Set this apparatus up in a fume hood in which low pressure gaseous nitrogen is available for purging excess N_2O_4 from the 1/16 inch I.D polytetrafluoroethylene tubing. The flow-through spectrophotometric cell depicted shall conform to the following:

a. 2.00 ± 0.02 mm sample path length.

b Coolant jacket for sample temperature control.

c Constructed of material compatible with N_2O_4.

d Constructed to fit instrument cell compartment

Figures 2 and 3 show suggested designs. Maintain the temperature of the flow-through cell at 32 ± 1.8°F (0 ± 1°C) by connecting the cell to a circulating ice-water glycol (20:1) bath located at the apparatus in the fume hood and/or at the spectrophotometer Maintain a purge of low pressure gaseous nitrogen, conforming to MIL-P-27401, in the spectrophotometer cell compartment or on the cell assembly to prevent condensation of moisture on the cell windows when the cell is in the spectrophotometer.

4.5 2 2 2 Procedure. Clamp a cylinder containing red-brown nitrogen tetroxide (NTO) in the apparatus according to Figure 1. Connect the spectrophotometric cell to the cylinder by means of 1/16 inch I.D polytetrafluoroethylene tubing. Purge the cell with dry nitrogen for several minutes. Close the inlet and outlet valves. Connect the cold bath to the cell and precool by pumping coolant for several minutes. Open the lower valve on the N_2O_4 cylinder slightly and control the sample flow with the cell valves. Pass several cell volumes (at least 10 ml) of sample through the cell Make sure the cell is free of bubbles, then close the cell outlet and inlet valves and the N_2O_4 cylinder valve, in that order. If possible, purge the sample line with nitrogen and disconnect the cell It is important to purge the cell with dry nitrogen before cooling, using several cell volumes of sample, and then checking for the absence of bubbles Place the cell in the spectrophotometer cell compartment and connect to an ice-water-glycol bath through suitable tubing. Allow the coolant to circulate for approximately 5 minutes. Set the absorbance to zero (100%T) at a wave length of 700 nm Remove the cell to the apparatus (Figure 1) in the hood and expel the N_2O_4 into the scrubber with a low pressure stream of nitrogen Repeat the above procedure substituting the

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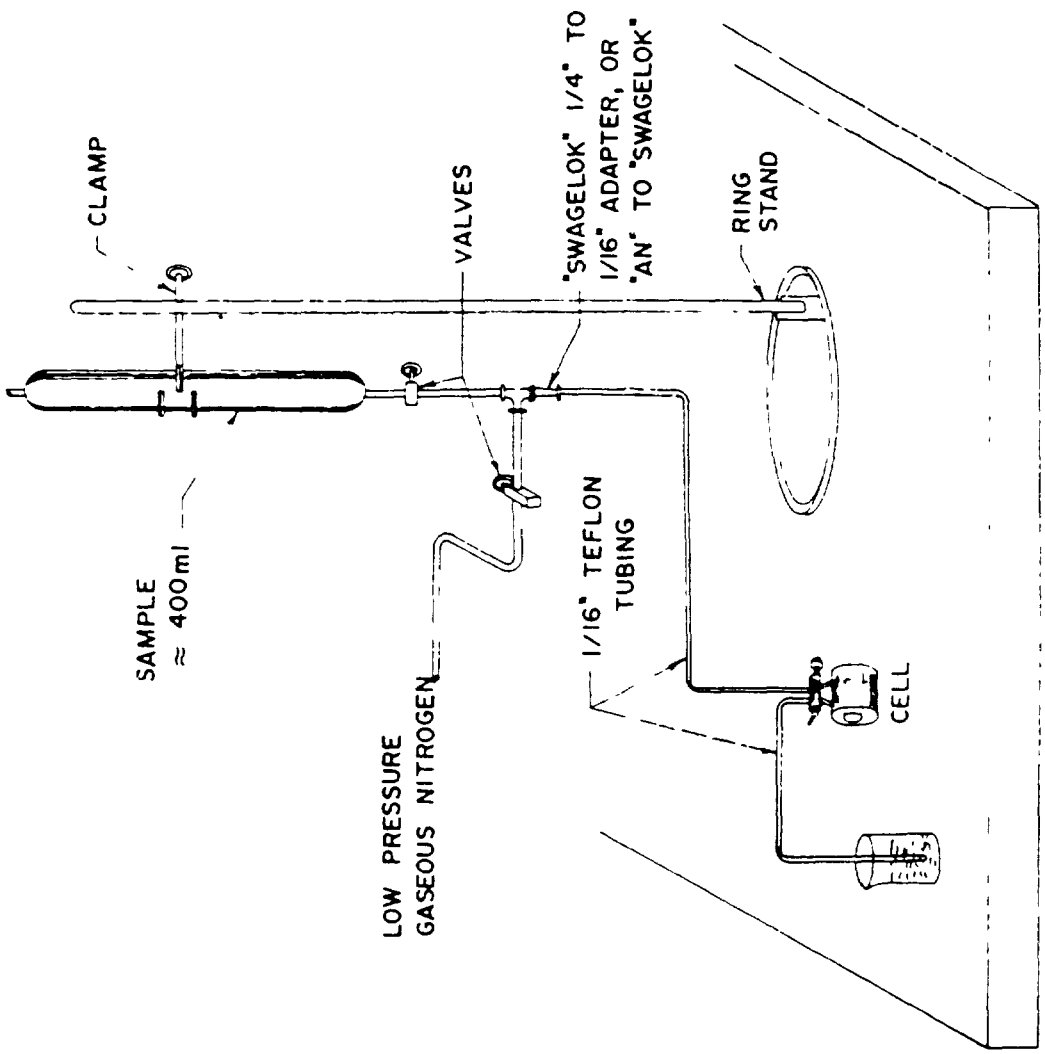


Figure 1. Nitric Oxide Sampler Apparatus

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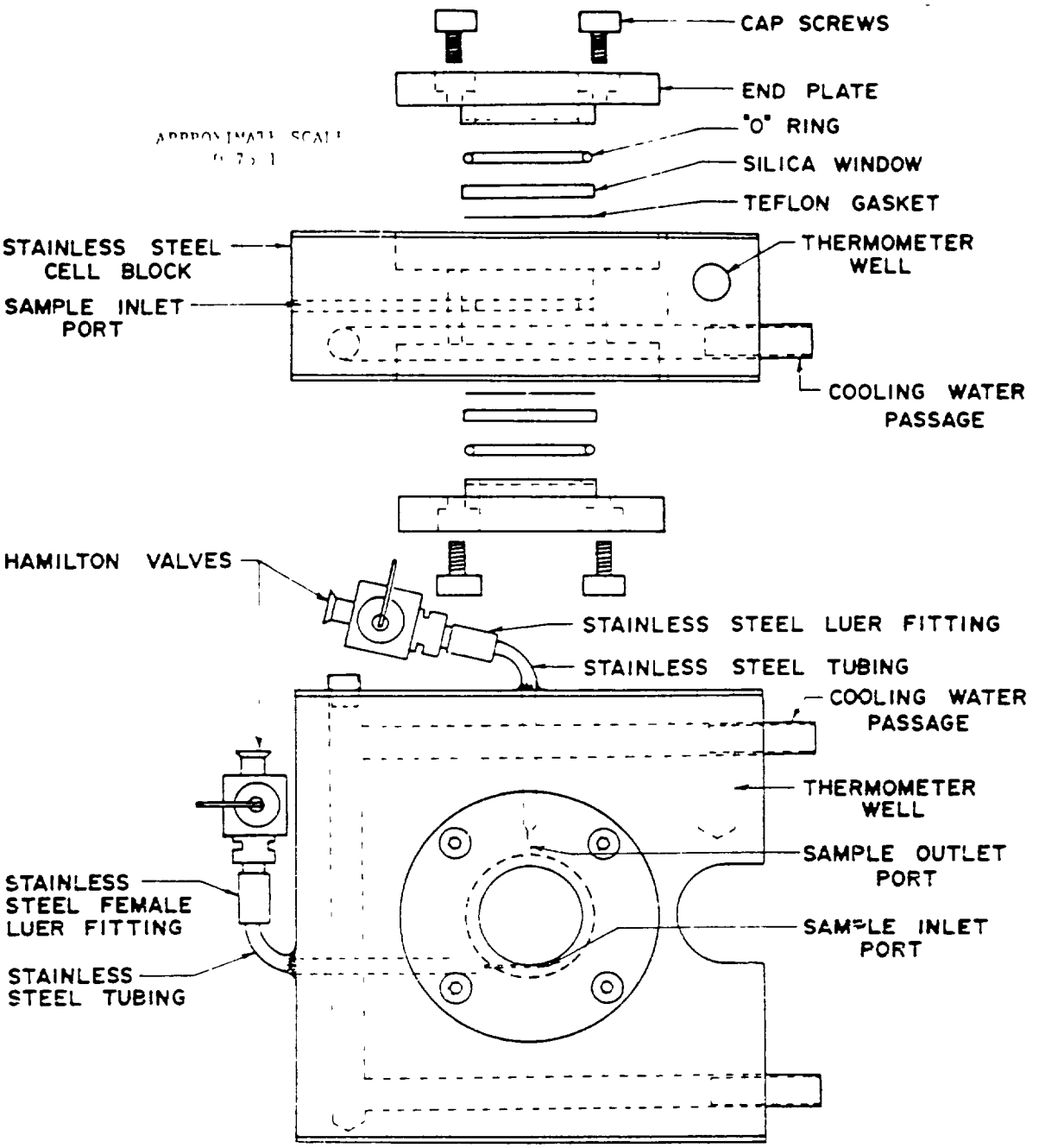


Figure 2. Stainless Steel Absorption Cell

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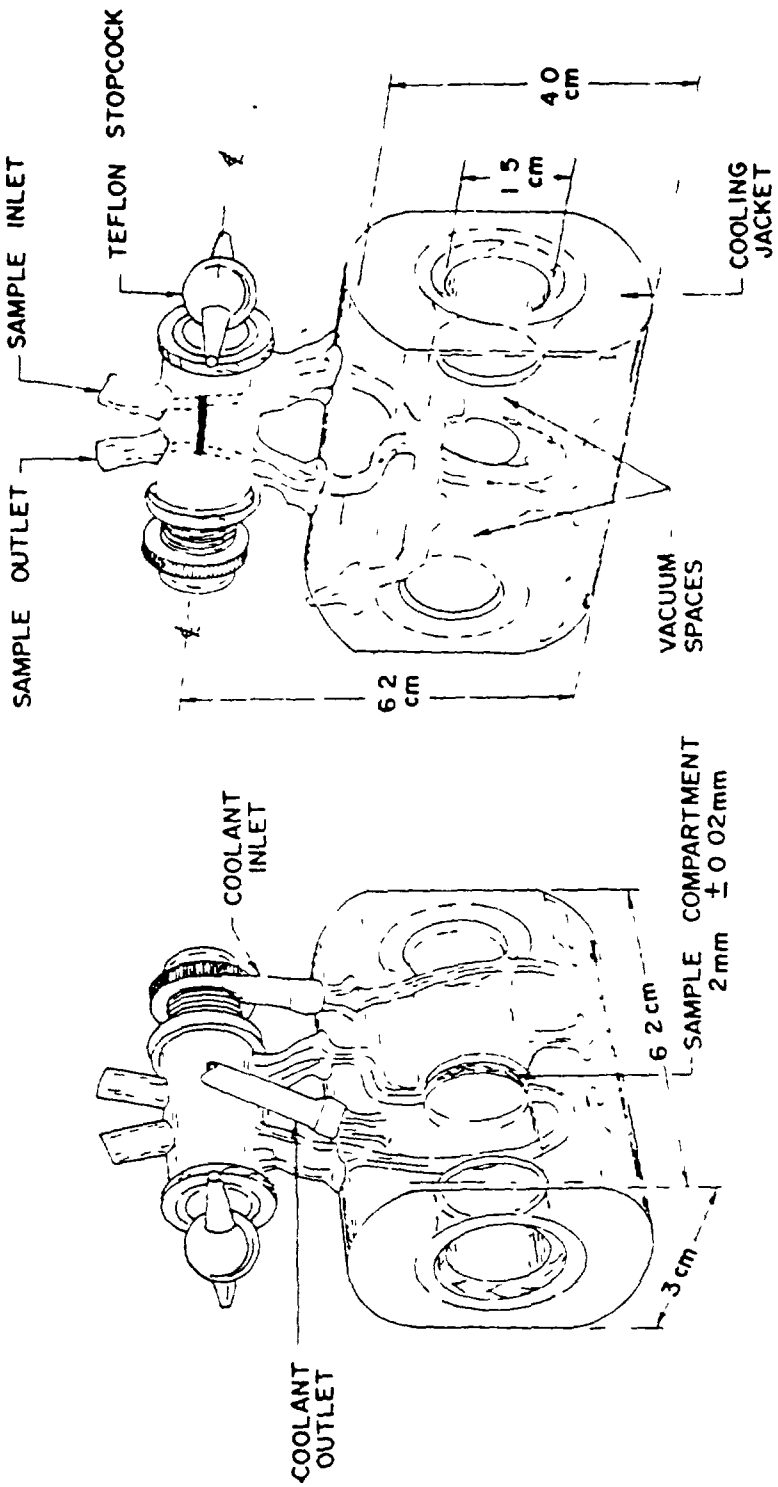


Figure 3. Glass Absorption Cell

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sampler. The temperature of the sampler should be 77 ± 3.6°F (25 ± 2°C). Measure the absorbance of the sample at 32 ± 1.8°F (0 ± 1°C) and at a wave length of 700 nm. Determine the NO percent by weight from a calibration curve prepared according to 4.5.2.2.3.

4.5.2.2.3 Calibration Curve A calibration curve shall be prepared as follows: Weigh accurately 58.164 g of nickel nitrate hexahydrate. $Ni(NO_3)_2$ $6H_2O$, to make 100 ml of a 2.0M stock solution. In a similar manner weigh out 26.008 g of sodium azide to make 100 ml of a 4M stock solution. Filter or decant the latter after allowing it to stand overnight. Standard A, equivalent to 0.5 wt% NO, is prepared by placing exactly 1.00 ml of the 2M Ni^{2+} solution in a 50 ml beaker Add 4 00 ml of the 4M NaN_3 solution and 15.0 ml distilled water and mix. Standard B, equivalent to 0.86 wt% NO, is prepared by mixing 1.00 ml of the 2M Ni^{2+} solution with 4 00 ml of the 4M NaN_3 solution, and 8.34 ml of distilled water in a second 50 ml beaker. Fill a 5.0 mm absorption cell with distilled water as a "blank" Set the spectrophotometer at 700 nm Set the blank at 100%T (0 absorbance). Determine the absorbance of Standards A and B in the same 5.0 mm cell at 700 nm. Plot the absorbance of Standards A and B versus the equivalent wt% NO (x-axis) and draw a straight line through the origin (0 wt% NO). The mixed synthetic standard solutions are stable for 48 hours. The sodium azide stock solution is stable for 30 days in the dark. The nickel nitrate stock solution is stable indefinitely

4.5 2 2 4 Reagents and Equipment The following reagents and equipment shall apply as test conditions of 4.5 2.2

a Reagents

(1) N_2O_4, pure or conforming to NTO of this specification

(2) Nitrogen gas, conforming to MIL-P-27401.

(3) Distilled water.

(4) Nickel(ous) nitrate hexahydrate, ACS reagent grade, or equivalent

(5) Sodium azide, practical, Eastman Cat. #P2352, or equivalent

(6) Sodium hydroxide, flakes, technical grade

(7) Ethylene glycol, technical grade.

(8) Oxygen, conforming to MIL-P-25508, Type I.

b Equipment

(1) Special N_2O_4 absorption cell 2 mm ± 0.02 mm

(2) 5 00 mm cell, silica, Pyrex, or Corex

(3) Ice bath.

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(4) Circulating pump

(5) Scrubber, 400 or 600 ml beaker, glass or plastic

(6) 1/16 inch polytetrafluoroethylene tubing, 12 inches.

(7) Ring stand, heavy duty, with clamp.

(8) Tubing, polyvinyl or rubber

(9) Fitting, 1/4 inch "T", AN or Swagelok.

(10) Fitting, 1/4 inch to 1/16 inch adaptor, AN to Swagelok

(11) Buret, 10 ml, micro (0.02 ml graduations)

(12) Valve, gas, 1/4 inch, AN or Swagelok

(13) Spectrophotometer, capable of furnishing narrow band energy at 700 nm and capable of accepting the special cell.

4.5.3 Nitrogen Tetroxide Assay. The N_2O_4 content of the sample shall be determined by the following procedure

4 5.3.1 Preparation of Sample.

4 5 3 1.1 Oxidation State Prior to the weighing and sealing of the assay sample, one of the following requirements shall be complied with

4 5 3 1.1 1 NTO composition NTO oxides of nitrogen shall be assayed without further chemical treatment. Proceed to paragraph 4.5.3 1.2 and then to 4 5.3 2 and 4 5.4

4.5 3 1 1.2 MON composition. The N_2O_4 assay of MON material shall be performed without further chemical treatment on the oxidized sample specified in paragraph 4.5 2.1 Proceed to paragraph 4.5.3.1 2 and to 4.5.3 2 If the NO content of the MON material was determined by the alternate method (spectrophotometric), then a 10-15 ml sample shall be oxidized as specified in paragraph 4 5.2.1 before proceeding to paragraph 4.5.3 1.2 and 4.5 3.2 below

4 5 3 1 2 Sample Sealing Cool the sample to approximately 32°F (0°C) with crushed ice Introduce the cooled sample into a clean, weighed (2 ml) glass ampoule directly from the sampler through a 1/16 inch stainless steel tube Insert the ampoule tip into an adequate size polyvinylchloride tube, approximately 6 inches long Freeze the contents of the ampoule by placing in a tray of crushed dry-ice or by some other appropriate method Lift the ampoule containing frozen N_2O_4 and wrap it with a piece of soft paper, and remove the polyvinylchloride tube. Seal the ampoule in the hot flame of a CH_4/O_2 or H_2/O_2 torch by rotating at an angle Test for leakage Wash, dry, and weigh the ampoule. Sample sizes should average approximately 1.3 g when 0 5N alkali is used in 4.5 3.2

4 5.3.2 Procedure Place the sealed ampoule into a heavy-walled glass stoppered Pyrex bottle, or iodine flask, containing 100 ml of water plus 20 ml

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of 30% hydrogen peroxide. Chill contents in ice water. Lubricate the stopper with stopcock grease and insert it into the ground glass neck of the bottle and/or flask Break the sealed ampoule by shaking vigorously, keeping the stopper firmly in place. Continue shaking until all brown fumes disappear. Return the bottle or flask to the ice water to keep the contents cool. Shake the bottle from time to time over a 15 minute period. Remove the stopper, rinse it off with distilled water into the contents of the bottle or flask, and replace it with an air condenser. Remove this assembly from the ice water and place on a steam bath or hot plate at moderate heat setting (approx 250°F/120°C) for 45 minutes. Allow the contents to cool to ambient temperature Rinse the air condenser carefully and remove it Pulverize the broken glass of the ampoule with the flat end of a heavy glass rod. Rinse the rod with distilled water into the bottle or flask. Add 3 - 5 drops of methyl red or methyl orange indicator and titrate with 0.5N NaOH solution to the yellow end point Run a reagent blank in the same manner as the sample

4.5.3.3 Calculation The wt% N_2O_4 shall be calculated by one of the following formulas

NTO

$$N_2O_4, \text{ percent by weight} = \frac{(a-b)N \times 4\ 6008}{W}$$

where.

a = ml NaOH for sample

b = ml NaOH for blank

N = normality of NaOH

W = grams of N_2O_4 in ampoule

MON composition

$$N_2O_4, \text{ percent by weight} = 1.534\ (A-B)$$

where.

$$A = \frac{(a-b)N \times (3\ 00 + 0.016 \times \%\ NO)}{W}$$

(see NTO for a, b, N definitions)

W = grams of oxidized N_2O_4 in ampoule

B = percent by weight NO

4.5.3.4 Reagents and Equipment. The following reagents and equipment shall apply as test conditions of 4.5.3.

a. Reagents

(1) Sodium hydroxide solution, 0.5N. Dissolve 20 g of ACS, or equivalent, reagent grade, low carbonate, NaOH in distilled water in a 1000 ml volumetric flask. Allow to

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cool and dilute to the mark with distilled water. Standardize this solution with Standard 0.5N HCl or Acid Phthalate to the methyl red or methyl orange end point Quantitatively transfer 30-50 ml of the standard acid into a beaker from a 100 ml buret. Titrate the acid with the 0.5N NaOH using the same buret after cleaning. Store the 0.5N NaOH in a polyethylene container and ensure the exclusion of carbon dioxide during storage.

(2) Distilled water.

(3) Methyl red or methyl orange indicator solution: 0.1 percent by weight in ethanol and distilled water.

(4) Hydrogen peroxide, ACS, or equivalent, reagent grade, 30 %wt

(5) Standard acid solution, 0 5N. 0.5N HCl standardized with primary standard grade diphenyl guanidine to the methyl red end point, or a certified standard grade HCl or acid phthalate.

(6) Diphenyl guanidine. Primary standard grade, F.G. Smith and Company, or equivalent

b Equipment

(1) Buret, 100 ml, 0.2 ml graduations.

(2) Flask iodine, 500 ml, heavy wall with 24/40 stopper

(3) Ampoule glass, 1 ml capacity.

(4) Flasks, volumetric, 1000 ml capacity.

(5) Ice bath.

(6) Low temperature bath, crushed dry-ice or equivalent.

(7) Glass rod.

(8) Glass air condenser, 24/40 ground glass end with drip tip

(9) Beaker, 200 to 400 ml, as available.

4.5.4 Water Equivalent. The water equivalent of the sample shall be determined by one of the following procedures. In the event of dispute, the procedure specified in paragraph 4.5.4.1 shall be the referee method.

4.5.4.1 Gas Chromatographic Method.

4.5.4.1.1 Analytical Apparatus. Assemble the analytical apparatus as shown in figure 4 Connect one side of a 2 µl stainless steel liquid inlet sampling valve to a 3 groove 6 port switching valve Connect the other side of the same 2 µl valve to a 12 inch x 3/8 inch Vycor glass tube filled with

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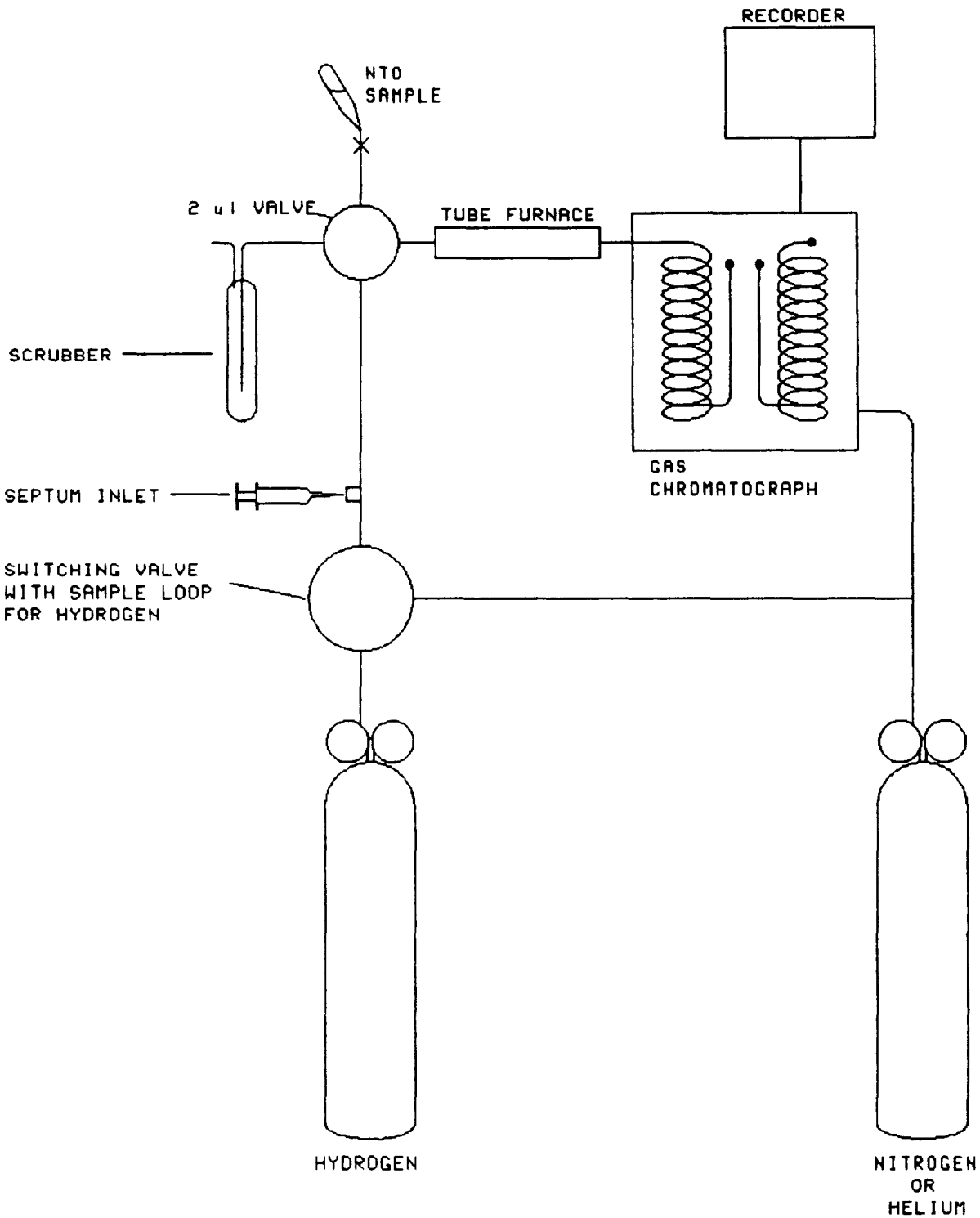


FIGURE 4. Water Equivalent Apparatus, Schematic

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copper filings and loosely plugged with copper turnings or quartz wool at each end The copper filings should be of sufficient surface area to produce 5 or 6 injections of N_2O_4 without peak deterioration but large enough in size to not restrict gas flow. These may be produced by a rough file on copper metal. Place the copper packed reactor in a split tube furnace capable of maintaining a constant temperature between 550 and 600°C. Connect the other end of the copper-packed reactor to the separation column of the gas chromatograph through 1/16th inch stainless steel tubing. This is the sample inlet for the separation. The distance between the copper reactor and gas chromatograph must be 2 inches or less and insulated with glass wool with an aluminum foil over-wrap. Do not heat interconnecting lines; insulation may be used where necessary. The inlet system should also contain a septum inlet or sample loop or loops of known volume (in the range of 10 to 100 µl) connected to the switching valve. This is used for calibration of the chromatograph. The column of the gas chromatograph should be of sufficient length of 1/8 inch tubing and contain a support material to adequately resolve water when nitrogen is used as the carrier gas The detector should have a minimum sensitivity of 0.25 millivolt per microgram of water. The reference and sample flow rate should be matched within 5 percent Optimize the column and detector temperatures to produce sharp peaks with adequate resolution To regenerate the reactor, engage the switching valve to the hydrogen gas source and isolate the chromatograph When the copper reactor is disconnected from the chromatograph for isolation turn off the detector to prevent air contact Regeneration is accomplished by alternating small amounts of hydrogen with the carrier gas flow into the apparatus until no more water can be detected at the outlet or when black copper oxide has returned to bright metallic copper Dead volume must be kept to a minimum throughout the system by using 1/16 inch OD tubing of the shortest length possible and filling voids in furnace connections with copper wool

4 5.4.1 2 Procedure The NTO sample specified in 4 5 3 1 1 shall be connected to the 2 µl liquid valve shown in figure 4 by means of a 1/8 inch stainless steel tube The MON material shall be oxidized according to 4 5.2 1 prior to connecting it to the 2 µl valve Excess oxygen in the vessel head space shall be released into the NTO scrubber and the vessel pressurized to 30 - 60 psi with dry nitrogen or helium. The apparatus specified in Figure 4 shall be in operational mode ready to analyze a 2 µl aliquot. When the vessel containing the oxidized NTO is connected to the 1/16 inch lines leading to the 2 µl valve the vessel should at no time be held directly vertical with the valve downward. Connect and clamp the vessel at a slight angle from vertical. It should also not be agitated prior to connection. These precautions are necessary to prevent any particulate from entering and plugging the 2 µl valve Allow the NTO to flow through the 2 µl sample valve to the NTO scrubber. When a steady stream of liquid flows introduce the aliquot and stop the NTO flow. Reject the first run after regeneration as it will be high due to removal of dead volume hydrogen. If a manual valve is used switching must be rapid and consistent with a low NTO flow to avoid valve leakage. Repeat the analysis 2 more times Determine the areas or peak heights. When the peak heights or areas of subsequent injections after the first rejected injection begin to decrease regenerate the copper in the reaction tube. Repeatability should be within 0 02% water equivalent If repeated injections are out of this range a dead volume check should be performed by checking all connections for proper filling of copper wool and the injection valve should be evaluated for leaking

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4.5.4.1.3 Calculation. The water equivalent shall be calculated as follows: Average the peak heights or areas and determine the micrograms (µg) of water from a previously prepared calibration graph (4.5 4.1 4) Calculate the water equivalent content using the following formula:

$$\text{Water equivalent, percent by weight} = \frac{\mu g\ H_2O}{29}$$

The peak heights or areas of the duplicates must be reproducible within 2 percent. If they are not, premature sample vaporization and/or a contaminated valve may be responsible for variations.

4.5.4.1.4 Calibration. A calibration curve shall be prepared as follows: Using a pressure-lok gas syringe having a 0 - 50 µl range, inject a 5 µl sample of hydrogen (at ambient pressure) into the partially oxidized copper reactor and record the peak height and area for the resultant water peak Repeat this using 10 µl, 25 µl and 50 µl of hydrogen gas. Calibration may also be performed by using a sample loop of known volume connected to a switching valve. The volume may be determined by the difference in weight when filling the loop with a solvent of known density. Record peak heights and areas for each sample. The calculation to convert µl hydrogen for each sample to µg water is as follows:

$$\mu g\ H_2O = \frac{18(a \times P \times 273)}{(22.4 \times 760 \times T)} = \frac{0.2887\ (aP)}{T}$$

where

a = µl of each sample.

P = ambient barometric pressure in mm Hg

T = ambient temperature in °K

Construct a calibration graph of peak height or area versus µg H_2O and draw a smooth line through the origin. A single point calibration may be used using one sample loop to confirm the calibration in subsequent analyses. This evaluates the detector or column deterioration while preventing the leaking of syringe ports or connections to sample loops deteriorated from multiple removal. The single point calibration should be performed before and after each set of samples.

4.5.4.1.5 Reagents and Equipment. The following reagents and equipment shall apply as test conditions of 4.5.4.1.

a. Reagents

(1) Hydrogen, cylinder with regulator. Prepurified grade gas, 99.95%, 20 ppm O_2 maximum.

(2) Carrier gas. nitrogen conforming to MIL-P-27401, or helium conforming to MIL-P-27407.

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(3) Copper, 20 - 40 mesh: Pure, ACS reagent grade chips or filings, "cuprin", or equivalent, size may be optimized for the particular system to obtain 5 - 6 injections per regeneration.

(4) Copper turnings: Pure, ACS reagent grade, or equivalent Copper wool or copper pot and pan scrubber is sufficient.

(5) Gas chromatographic column: Acceptable columns and conditions as follows:

(a) 6 to 10 ft of 5% polyethyleneglycol on "Teflon" 40 - 60 mesh, at 158°F (70°C) and 25 to 30 cc/min flow rate

(b) 6 to 10 ft of polystyrene-polyvinyl bead absorbent, "Porapack QS", or equivalent, at 212°F (100°C) and 25 to 30 cc/min flow rate

(c) Polyhalocarbon supports, "Kel F", or equivalent, with polyesters or polyglycols are satisfactory

(d) Diatomaceous earth supports or other water absorbing supports shall not be used as support materials

(e) Column shall be aluminum or stainless steel, 1/8 inch OD

b Equipment

(1) Tubing. Stainless steel 1/8 and 1/16 inch OD as required in figure 4

(2) Switching valve: H_2-N_2 supply, Carle No 2821, or equivalent.

(3) 2 µl liquid injection valve: internal construction of stainless steel with polytetrafluoroethylene seals, Carle No 2015 or equivalent

(4) N_2O_4 scrubber N_2O_4 disposal shall be accomplished in accordance with applicable local anti-pollution laws

(5) Gas chromatograph. A chromatograph with a thermal conductivity detector

(6) Recorder A potentiometric recorder with a 1 millivolt span and a 1 sec FS response.

(7) Tube furnace. 10 to 12 inches long with a 3/4 to 1-1/4 inch bore. Switch settings (voltage) shall be determined with a pyrometer so that it shall operate between 1022 to 1112°F (550 to 600°C), 500 to 1000 watts is adequate.

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(8) Precision gas syringe. 0 - 50 µl capacity, graduated in 0.5 µl. Precision Sampling corporation, No 050024 with needle, or equivalent

(9) Reactor tubing. Vycor glass, 12 inch x 3/8 inch corning No. 743180, or equivalent.

(10) Alternate calibration sample loops: Stainless steel sample loop switching valve 1/16 inch diameter, 10 to 100 µl in volume.

4.5.4 2 Near Infrared Absorption Method.

4.5.4.2.1 Preparation Set the conditions of a Lamda 9 spectrophotometer as specified in TABLE III. These are suggested parameters and they may be varied by the analyst to optimize the procedure. Set the temperature of a circulating low temperature Freon TF bath, or equivalent, to 21.2°F ± 1.8°F (-6°C ± 1°C). Attach the cell to the cooling bath and start the circulation. Purge the sample compartment with nitrogen so that the cells do not become coated with moisture

TABLE III. Conditions for Lamda 9 Spectrophotometer

	Method	Scan/Manual
01	Ord/Absc Mode	Abs/WL
02	Slit/Gain/Sens	2.00 nm/ /2
03	Scan Speed	240nm/min
04	Response	0.5 S
05	Lamp/Det	319.2 / 860.8 nm
06	Cycles/Time	1 / 0.05 min
07	Peak Threshold	0 01 A
08	Recorder	SERIAL / DASH 1
09	Ord Min/Max	.0000 / 0.500
10	Absc Min/Max	1300 0 / 1650.0
11	Abscessa Format	20 nm/cm
12	Printer	/ SCALE /

4 5 4 2 2 Procedure Transfer some of the oxidized N_2O_4 to a small cooled Erlenmeyer flask. Load the cell from the flask, using a cooled 2 ml syringe, then place the loaded cell in sample beam of the spectrophotometer Place an empty cell in the reference beam. Set the spectrophotometer to 1300 nm, then zero the instrument. Push the RUN/STOP button to start the scan The absorption of the HNO_3 band pass has to be measured from the baseline which is found by connecting the minimal points at 1430 nm and 1615 nm The band pass of atmospherical (free) water can be found by entering 1404 PEAK FIND. Record the absorbance.

4.5.4.2.3 Calculations. The wt% water equivalent shall be calculated by one of the following formulas.

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NTO composition

$$\text{Water Equivalent (wt\%)} = \frac{A_{HNO_3}}{hbf} + \frac{A_{H_2O}}{ab}$$

MON composition

$$\text{Water Equivalent (wt\%)} = \left[\frac{A_{HNO_3}}{hbf} + \frac{A_{H_2O}}{ab}\right]\left[\frac{100 + C(\%\ NO)}{100}\right]$$

where:

A_{HNO_3} = Absorption of HNO_3

A_{H_2O} = Absorption of free water

h = Absorption coefficient of HNO_3 = 0 178

b = Thickness of sample (cm)

f = Factor for conversion of HNO_3 to H_2O = 7

a = Absorption coefficient of free water = 0.55

C = Calculation coefficient (NO to N_2O_4) = 0.534

4 5.4.2.4 Reagents and Equipment The following reagents and equipment shall apply as test conditions of 4 5.4.2

a. Nitrogen. conforming to MIL-P-27401, Type 1

b Circulating low temperature Freon TF bath, or equivalent

c. Two 1 cm cells with cooling jackets

d Syringe, 2 ml

e Erlenmeyer flask, 50 ml

f. Spectrophotometer, Lambda 9

4.5 5 Chloride Content. The chloride content of N_2O_4 is determined by one of the following methods In the event of a dispute the procedure specified in paragraph 4 5.5.1 shall be used as the referee method

4.5.5.1 Silver Nitrate Titration Method.

4.5.5 1 1 Sample Preparation Place a 250 ml beaker on a magnetic stirrer in a laboratory hood Add to the beaker a polytetrafluoroethylene stirring bar and 150 ml 1M NaOH solution. Activate stirrer Weigh a sampler containing NTO or oxidized (4.5.2.1) MON material to 0.1 gram A sample

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vessel that holds approximately 100 ml is best for weighing accuracy. Attach a polytetrafluoroethylene tube to the sampler. Slowly add 6 to 8 ml (9 to 12 grams) of sample, via the tube below the surface of the caustic. The rate should be low enough so that no red vapors form from breaking bubbles which will reduce the amount of sample introduced. Remove the tube and determine the sample weight by difference. Slowly add with stirring an amount of urea equal to one-half (0.5) the sample weight in grams. Care should be taken to prevent sample loss due to bubbling. Slowly add with stirring 1:1 HNO_3 until gas evolution ceases or a pH of 2.5 to 3.0 is reached. The pH of the solution should be checked often while adding the acid to make sure that a pH of 2.5 is not exceeded. Allow the solution to stand 15 minutes in preparation for titration

4.5.5 1.2 Procedure. Fill a syringe pump, adjusted to deliver a constant volume of titrant/unit time (in the range of 0.5 to 0.7 ml/min.), with 0.05M $AgNO_3$ solution. Connect the pump to the prepared sample with a polytetrafluoroethylene tube Place a chloride or silver standard ion electrode, and a double junction reference electrode in the sample solution. The electrode should be connected to a millivolt meter and recorder. With sample stirring, turn on meter, recorder, and pump. Chart speed should be approximately one inch per minute. Mark the point where the syringe pump delivery begins As soon as the 'S' shape curve is completed, turn off the pump and recorder. Determine the inflection point by drawing tangents to the curve, dropping a perpendicular from the upper to lower tangent, and constructing a perpendicular bisector. Measure the exact titrant delivery time, such as 5.75 minutes. Run a blank in the same manner. Delivery time of titrant in minutes multiplied by delivery rate in ml/min equals titrant volume in ml. Alternately the 'S' shape may be generated by incremental additions of the silver nitrate solution with a 25 ml burette and detection of millivolts from the millivolt meter The millivolts vs milliliters shall be plotted during silver nitrate addition to determine when the 'S' curve is approaching. Increments should be 1 ml or less to the inflection point and 0.25 or less at the rise to the inflection point, and returning to 1 ml increments after the rise. An automatic titrator may also be substituted for the syringe pump, provided the detection uses the same electrodes and the end point is identical to the plotted 'S' curve. The silver nitrate may require dilution for automatic titrators

4.5.5.1.3 Calculation. The percent by weight chloride is calculated as follows:

$$\text{Chloride, percent by weight} = \frac{(\text{ml } AgNO_3 - \text{ml blank}) \times M \times 35.45 \times 100}{W}$$

where. M = Molarity of $AgNO_3$ titrant.

W = sample weight in mg

4 5.5.1 4 Calibration. The $AgNO_3$ molarity (M) is obtained by titrating, in triplicate, a standard sodium chloride solution by the procedure in 4 5.5 1.2.

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4.5.5.1.5 Reagents and Equipment. The following reagents and equipment shall apply as test conditions of 4.5.5.1

a. Reagents

(1) Sodium chloride ACS reagent grade, or equivalent, dried for one hour at 212°F (100°C).

(2) Standard sodium chloride solution, 1 ml= 1.00 mg Cl. Weigh 1 648 g dry sodium chloride into a one liter volumetric flask Dilute to the mark with distilled water. Transfer the well mixed solution to a clean polyethylene bottle for storage.

(3) NaOH solution, 1M: Dissolve 40 g ACS reagent grade, or equivalent, sodium hydroxide in distilled water Dilute to 1000 ml.

(4) HNO_3 solution, 1.1· Mix 500 ml ACS reagent grade, or equivalent, nitric acid with 500 ml distilled water

(5) $AgNO_3$, 0 05M Dissolve 8.494 g ACS reagent grade, or equivalent, silver nitrate in distilled water to which has been added 10 ml of 1 1 HNO_3 solution Dilute to 1000 ml Store the well mixed solution in a light-proof bottle

(6) Urea ACS reagent grade, or equivalent, crystals

b Equipment

(1) Balance, analytical, 0.1 g sensitivity

(2) Chloride ion selective electrode. Orion No. 941700, or equivalent

(3) Silver/sulfide ion selective electrode. Orion No. 941600, or equivalent

(4) Double junction reference electrode: Orion No 900200, or equivalent

(5) Millivolt meter, high impedance pH meter with recorder output. Orion Model 701A, or equivalent

(6) Recorder, 1, 10, 100 mV full scale. Beckman Model 1005, or equivalent.

(7) Syringe pump, 50 ml syringe, delivery rate 0 1 to 1.0 ml/min· Sage Model 341-A or equivalent, with syringe

(8) Magnetic Stirrer.

(9) Polytetrafluoroethylene tubing and coated stirring bars

(10) Glassware, assorted. Pyrex, or equivalent

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4.5.5.2 Colorimetric Method.

4.5 5.2.1 Procedure Weigh the cylinder containing the oxidized sample prepared in 4.5.2.1 to the nearest 0.10 gram. Slowly bubble 1 to 50 g of N_2O_4 through 40 to 50 ml of chloride-free distilled water contained in each of two 250 ml Fleming absorbers, connected in series with polyvinyl tubing The sample must be scaled to contain less than 0.050 mg Cl^- Combine the two solutions in a 250 ml beaker Prepare a reagent blank with 50 ml of chloride free distilled water in a second beaker. Reweigh the cylinder. Add 2 to 5 drops of 2,4-dinitrophenol to each of the 250 ml beakers and neutralize each with concentrated ammonia to the yellow end-point. Adjust the volume in the sample beaker to contain one gram of N_2O_4 per milliliter of solution by evaporating on a hot plate Into three other 25 ml volumetric flasks add 0.010 mg Cl^-, 0 020 mg Cl^-, and 0.040 mg Cl^- of the dilute standard Cl^- solution, using a 5 ml microburet. Add chloride free distilled water to all flasks so that they contain about 20 ml of liquid. Add exactly 2.0 ml of the iron (III) ammonium sulfate reagent to the flasks and mix. Add exactly 1 0 ml of the saturated mercury (II) thiocyanate reagent, make up to mark with chloride free distilled water and mix. Place all flasks in darkness for 15 to 30 minutes. Measure the absorbance of the sample and the standards in a 5.00 cm cell at 460 nm, after setting the blank in a 5.00 cm cell at zero absorbance (100%T) Plot the absorbance of the standards versus mg Cl^- and draw a curve through the origin.

4 5.5.2 2 Calculation The weight percent chloride shall be calculated as follows·

$$\text{Chloride, percent by weight} = \frac{(\text{mg } Cl^- \text{ from curve}) \times F \times 100}{\text{mg } N_2O_4}$$

where. F = aliquot factor (i.e 1/1 to 5/1)

4 5 5.2 3 Reagents and Equipment The following reagents and equipment shall apply as test conditions of 4.5 5.2.

a Reagents

(1) Chloride free distilled water. The water is considered chloride free if a 100 ml sample, treated with silver nitrate, shows no turbidity during a 24 hour period, when stored in darkness.

(2) 2,4-dinitrophenol indicator, 0 1 %wt in alcohol and water.

(3) Ammonia, ACS reagent grade, 28% NH_3

(4) Nitric acid· ACS reagent grade, 70% HNO_3

(5) Iron (III) ammonium sulfate, 0 25M· Weigh 60.275g of iron (III) ammonium sulfate dodecahydrate, $FeNH_4(SO_4)_2\ 12H_2O$, ACS reagent grade, and make up to 500 ml volume with 9M nitric acid. Filter or decant after 24 hours

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(6) Saturated mercury (II) thiocyanate Dissolve 100 mg ACS reagent grade $Hg(SCN)_2$ in 100 ml 95% ethanol Allow to stand, decant, and store in darkness

(7) Stock standard chloride solution. Dissolve 1 649 g of dry primary standard grade sodium chloride in chloride free distilled water, and dilute to the mark in a 500 ml volumetric flask (1 ml = 2 0 mg Cl^-).

(8) Dilute standard Cl^- solution· Dilute 1 00 ml of the stock standard chloride solution to the mark in a 200 ml volumetric flask with chloride free distilled water (1 ml = 0 010 mg Cl^-).

b Equipment

(1) Chloride free glassware: All glassware used must be washed with nitric acid and rinsed with chloride free distilled water.

(2) Volumetric flasks, 25 ml, 100 ml, 200 ml, 500 ml.

(3) Volumetric pipette, as required

(4) Beakers, 250 ml.

(5) Analytical balance, 0-100 g, 0.1 mg sensitivity

(6) 5.00 cm spectrophotometer cells

(7) Spectrophotometer, capable of furnishing narrow band light between 400 to 500 nm, and capable of accepting 5 0 cm cells

(8) Oven, variable temperature, to 230°F (110°C)

(9) Ringstand and clamps, as required.

(10) Microburet, 5 00 ml capacity

(11) Hot plate

(12) Fleming gas absorbers, 250 ml size.

4.5.6 Non-Volatile Residue. The non-volatile residue of the propellant shall be determined by the following method.

4 5 6 1 Procedure Thoroughly clean two platinum evaporating dishes per sample Dry the dishes in an oven at 220 ± 10°F (105 ± 5°C) After drying cover the dishes with a watch glass and allow them to cool next to the balance for 30 minutes. Weigh each dish to the nearest 0 01 milligram Place a hot plate in an exhaust hood and place the weighed dishes on it Add 1 ml concentrated H_2SO_4 to each dish Transfer 50 ml of the N_2O_4 to each dish Using the hot plate, evaporate the sample until only 1 to 2 ml remain in each dish Oven dry the dishes for one hour beyond the point when no liquid

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remains in each dish. After drying cover the dishes with a watch glass and allow to cool next to the balance for 30 minutes. Weigh each dish to the nearest .01 mg

4.5.6.2 Calculation. The %wt non-volatile residue shall be calculated by the following formula:

$$NVR\ (\%\ wt) = \frac{(W_R - W_E)}{V * D}$$

where.

W_R = Dish weight with residue (grams)

W_E = Dish weight empty (grams)

V = Volume of sample used (liters)

D = Density of propellant at measured temperature

4 5 6.3 Reagents and Equipment. The following reagents and equipment shall apply as test conditions of 4.5.6.

a Platinum evaporating dishes.

b. Hot plate, porcelain topped.

c. Graduated cylinder, 50 ml.

d Oven, 230°F (110°C).

e Analytical balance; .01 mg sensitivity

f. Watch glasses.

g Sulfuric acid, reagent grade

4 5.7 Iron Content The iron content of the low-iron grade MON-1 and the MON-3 is determined as follows:

4 5 7 1 Procedure. Using a 100 ml burette transfer 20 to 50 ml of the low-iron grade MON-1 or MON-3 to a 100 ml beaker. Use larger sample sizes when less iron is expected. Calculate the weight of sample by using the appropriate density. Place the beaker on a hot plate containing an asbestos pad or equivalent and set to the low heat range. Slowly evaporate the sample Add 1 ml of concentrated H_2SO_4 when the sample evaporates to approximately 5 ml Allow the sample to evaporate to a clear-straw colored liquid, then set it aside to cool to room temperature. Transfer the contents of the beaker to a 25 ml volumetric flask Rinse the beaker with additional distilled water and transfer the rinse water to the 25 ml volumetric flask Dilute the sample in the flask to the mark with distilled water Stopper the flask and shake to ensure complete mixing Determine the absorption of the sample in the volumetric flask at an iron wavelength closest to 2480 Å on a suitable atomic absorption spectrophotometer after setting the zero absorption with distilled water Prepare a series of iron standard solutions in the 1 to 10 ppm range

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in accordance with 4 5.7 3. Determine the absorption of the iron standards at the iron wavelength.

4 5 7.2 Calculation The iron content of the low-iron grade MON-1 or MON-3 shall be calculated as follows:

$$I = \frac{(A_{MON} - A_{blk})CVF_d}{A_o W}$$

where.

A_{MON} = Absorbance of MON sample

A_{blk} = Absorbance of blank solution prepared without sample

A_o = Absorbance of nearest standard solution

C = Concentration of nearest standard solution

V = Volume of the undiluted sample solution (ml)

F_d = Dilution factor, if used

W = Weight of N2O4 used (g)

I = Iron concentration (ppm by wt)

4.5 7 3 Calibration. Prepare 1 ppm Fe, 2 ppm Fe, 5 ppm Fe, and 10 ppm Fe standard solutions by the separate addition of 0.2 ml, 0.4 ml, 1.0 ml and 2 ml aliquots of the 1000 ppm Fe stock standard solution, to each of four 200 ml volumetric flasks To each flask are added 50 ml of distilled water, and 2 ml of concentrated sulfuric acid Dilute to the mark with distilled water Transfer each of the four standard solutions to plastic screw-cap bottles. Determine the absorption of the four standards at 2480 A, which is the same Fe line as for the sample, using distilled water to set the instrument at zero absorption The standard iron solution having the absorption nearest to the sample is used to calculate the iron content of the sample.

4 5 7 4 Reagents and Equipment The following reagents and equipment shall apply as test conditions of 4 5 7

a Reagents

(1) Sulfuric acid, reagent grade.

(2) Hydrochloric acid, reagent grade

(3) Iron wire, reagent grade

(4) Iron standard solution, 1000 ppm Use a commercial iron standard or prepare the standard by the following method Weigh 1.000 g of reagent grade iron wire accurately. Transfer to a 1000 ml volumetric flask and add 15 ml of concentrated HCl solution. Heat if necessary to completely

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dissolve the iron wire. Dilute to the mark with distilled water.

(5) Distilled water.

b Equipment

(1) Volumetric flasks, 10, 25, 1000 ml, as required

(2) Beaker, 50 ml.

(3) Cylinder, graduated, 10 ml.

(4) Balance, triple beam, 5 lb capacity.

(5) Balance, macro, 200 g capacity, Mettler, or equivalent.

(6) Hot plate, controllable heat settings.

(7) Plastic bottles, screw-cap, 8 oz., as required

(8) Atomic absorption spectrophotometer, with an Fe hollow cathode lamp

4 5.8 Particulate The particulate matter of N_2O_4 is determined as follows:

4.5.8 1 Preparation of Test and Control Membrane Filters Two 47 mm membrane filters of nominal pore size (10µm) are required: a test and a control membrane filter Matched-weight membrane filters may be used if so desired If matched-weight membrane filters are used, it is unnecessary to carry out the procedures detailed in this section because they had been carried out previously by the membrane filter supplier. The two membrane filters used for each individual test should be identified by marking the petri dishes used as containers Glassware used in preparation of membrane filters must be cleaned as described in 4.5.8.2. Using forceps place the test and control membrane filters in clean petri dishes. Place the petri dishes, with lids slightly ajar, in an oven at 194 ± 9°F (90 ± 5°C) and leave them for 30 min Remove the petri dishes from the oven and place them near the balance The petri dish covers should be ajar but still protecting the membrane filters from contamination from the atmosphere. Allow 30 min for the membrane filters to come to equilibrium with the ambient air temperature and humidity Remove the control membrane filter from the petri dish with forceps, handling by the edge only, and place it centrally on the weighing pan Weigh it and return it to the petri dish Repeat for the test membrane filter.

4.5 8 2. Preparation of Apparatus The following procedure should be followed, utilizing an apparatus similar to that shown in Figure 5 An alternate apparatus may be used provided that it achieves the same end. For example, a high pressure disc filter assembly may be used. Wash the petri dishes with warm water containing detergent Then rinse with warm water and finally with distilled water. Rinse thoroughly with filtered isopropyl alcohol Rinse thoroughly with filtered trichlorotrifluoroethane (Refrigerant-113). Drain for a few seconds, then air or oven dry Ensure

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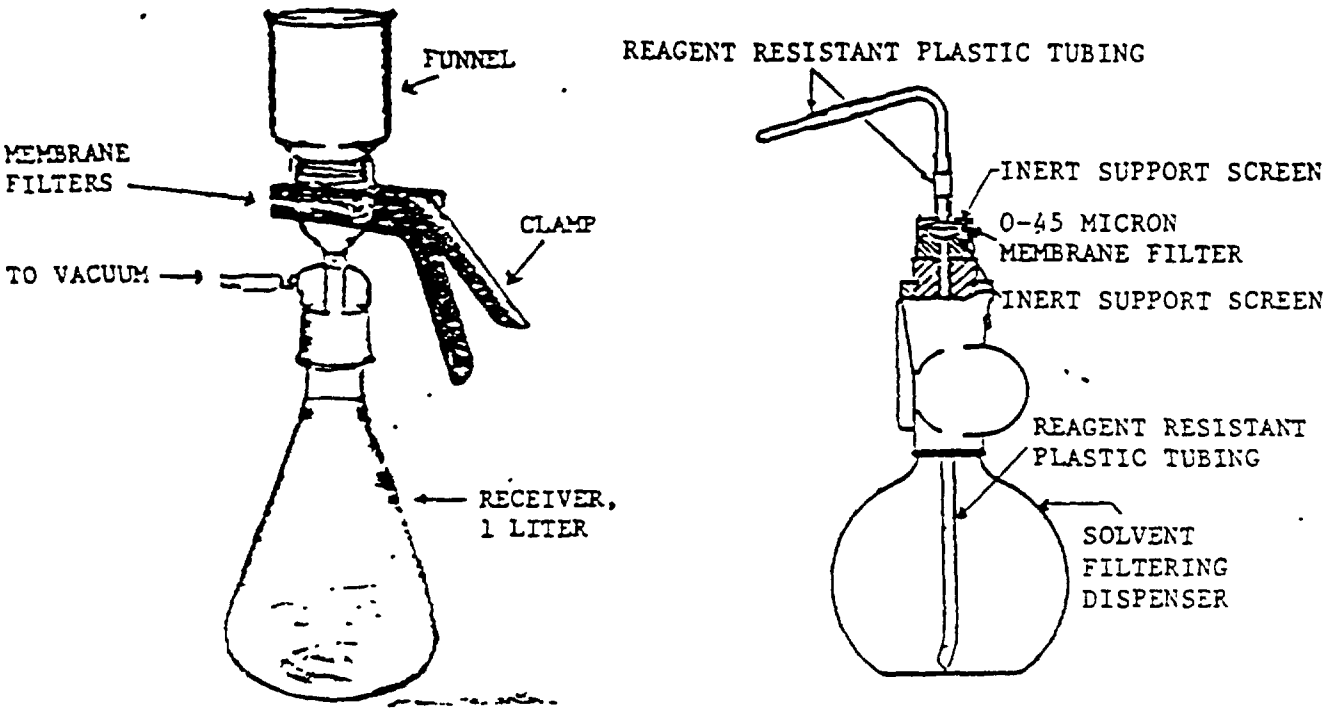


Figure 5. Apparatus for Determining Particulate

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that all glass and plastic tubing attached to the solvent filtering dispenser is clean by flushing thoroughly with filtered trichlorotrifluoroethane (Refrigerant-113)

4.5.8.3 Sampling. When possible, 400 - 500 ml (1 pt) of sample should be passed through the filter assembly during sampling The sample volume actually employed must be reported

4.5.8.4 Procedure Assemble the type of apparatus shown in Figure 5. Place the tip of the delivery spout of the solvent filtering dispenser in direct contact with the top membrane filter Introduce trichlorotrifluoroethane (Refrigerant-113) until the reservoir is full Apply vacuum to the flask and allow approximately 250 ml of trichlorotrifluoroethane (Refrigerant-113) to pass from the solvent filtering dispenser through the membranes and into the vacuum flask. Immediately after, allow a volume of 400-500 ml (1 pt) of sample to pass through the membranes and into the vacuum flask. When fuming ceases, slowly release the vacuum Remove the membranes and reservoir above the vacuum flask and carefully dismantle in an upright position Carefully remove the test and control membrane filters, and place them in clean, covered petri dishes Dry and reweigh the membrane filters taking care not to disturb the contaminant on the surface of the test membrane filter. The drying procedure described in 4.5 8.1 should be followed in detail.

4.5.8 5 Calculation Subtract the initial weight of the test membrane filter, W1, from the final weight, W2. Subtract the initial weight of the control membrane filter, W3, from the final weight, W4. Divide the corrected weight of contaminant (W2 - W1) - (W4 - W3) by the volume of sample filtered and report the result as total contaminant, expressed in milligrams per liter. If matched-weight membrane filters have been used for the test, then W1 = W3 and the corrected weight of contaminant becomes W2 - W4. Report the result to the nearest 0 01 mg/liter, and also the sample volume used in the test.

4 5 8 6 Reagents and Equipment. The following reagents and equipment shall apply as test conditions of 4 5.8

a. Reagents

(1) Isopropyl alcohol, reagent grade

(2) Filtered trichlorotrifluoroethane

(3) Distilled water.

(4) Liquid Detergent, water-soluble

(5) Refrigerant-113 (trichlorotrifluoroethane).

b Equipment

(1) Analytical balance, single or double pan, the precision standard deviation of which must be 0.07 mg or better

(2) Oven, of the static type (without fan-assisted air circulation), controlling to 194 ± 9°F (90 ± 5°C)

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(3) Petri dishes, approximately 50 mm in diameter with removable glass supports for membrane filters.

(4) Forceps, flat-bladed with unserrated, non-pointed tips.

(5) Vacuum system.

(6) Test membrane filters, 47 mm diameter, polytetrafluoroethylene, filter disc with rating of 10 ± 3µm

(7) Control membrane filters, 47 mm diameter, polytetrafluoroethylene, filter disc with rating of 10 ± 3µm.

(8) Solvent filtering dispensers, 0.45 µm membrane filters to be provided in the delivery line (see Figure 5).

(9) Filter apparatus, all-glass, 47 mm, according to Figure 5.

(10) Air ionizer, for the balance case.

(11) Disc filter holder, in-line, reagent resistant plastic, 47 mm.

4.6 Preparation for Delivery Inspection. The packaging and marking of the propellant shall be inspected to determine compliance with the requirements of Section 5.0 of this specification.

5.0 PREPARATION FOR DELIVERY

5.1 Packaging. The product furnished under this specification is a hazardous material as defined and regulated by the Department of Transportation (DOT) regulations All packaging to be shipped commercially by any mode of transportation shall comply with the requirements of DOT regulations 49 CFR 171-190 and United Nations (UN) directives for international shipments applicable to the mode of transportation DOD may obtain from DOT an exemption in accordance with 49 CFR 170.13 through the Commander, Headquarters Military Traffic Management and Terminal service, Attn. Safety Division (TES), Washington, D C 20315 The product shall be packaged in containers and unit quantities as specified by the procuring activity (6.2 and 6.3).

5 2 Preparation of Containers. Prior to filling, the contractor shall determine the condition of all containers to insure that they are free from contamination and suitable for shipment and storage. Contractor owned containers shall be cleaned and repaired by the contractor at his own expense. Leased or Government owned containers shall be cleaned and repaired in accordance with the schedule established in the contract or purchase order

5.2.1 Cleaning and Repair. To the extent which is practical, all containers shall be visually inspected internally and externally for the presence of water, rust, scale, oil film, leakage or other foreign matter, and physical damage. Any physical damage which would endanger safe transportation of the propellant shall be repaired prior to reuse. If evidence of internal contamination is found, the container shall be recleaned by a suitable method to remove the contamination.

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5.2.2 Testing. Containers shall be hydrostatically or otherwise tested at a frequency specified by and in accordance with DOT requirements or as otherwise specified by the procuring activity.

5 2 3 Gaskets. Gaskets used to seal container openings shall be polytetrafluoroethylene or other material compatible with the propellant and approved for use by the procuring activity. The contractor shall assure that all gaskets are serviceable and furnish new gaskets when necessary so that a tight seal is assured.

5.3 Filling. The filling by density, weight, pressure, temperature, or any combination thereof shall be in accordance with DOT requirements or as otherwise specified by the procuring activity. Containers for which the prescribed test have become due shall not be filled until retested.

5.4 Labeling and Marking. Each container shall be labeled, marked and placarded in accordance with MIL-STD-129, Code of Federal Regulation 49 (49 CFR), Department of Transportation (DOT) special permit, and United Nations requirements (if applicable). In addition, each container shall be marked 'INHALATION HAZARD'. An identification tag, precautionary label, and container color code with name identification shall be used on each cylinder.

5.4.1 Identification Tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climatic conditions shall be wired to the outlet port of the cylinder and shall contain the following information. propellant name, specification number with revision letter (MIL-P-26539D), designated composition, quantity, lot number, name of manufacturer, name of contractor (if different from manufacturer), and date of manufacture

5.4.2 Precautionary Label. A precautionary label prepared in accordance with ANSI Z129.1 shall be applied to each cylinder (6.6).

5 4 3 Container Color Code. Unless otherwise specified by the procuring activity, each cylinder shall be color coded in accordance with MIL-STD-101 The exact name identification to be marked on the outside of the cylinder shall be 'Nitrogen Tetroxide'. Any other name identification shall be obliterated by removing or overpainting.

6.0 NOTES

6.1 Intended Use The propellant described by this specification is intended for use as an oxidizer in rocket engines

6.2 Ordering Data.

6 2 1 Procurement Requirements.

a Title, number (MIL-P-26539D), composition, and date of this specification

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b. Method of shipment, type and capacity of containers, (5.1 and 6.3)

c. Quantity by weight in pounds (avoirdupois).

6.2.2 Contract Data Requirements. Data conforming to Data Item Description DI-T-3733, Quality Conformance Test Reports (Fuels), is a requirement for DOD deliveries in connection with this specification. The data item will be specified for delivery on the DD Form 1423.

6.3 Containers As of the date of this specification, the following listed containers are considered acceptable for military use and unless otherwise noted are approved by DOT regulations and special permits for use with nitrogen tetroxide.

a Sample quantities of product may be shipped in specification DOT 3A1800, 3AA1800, or 3E1800 cylinders as specified in 49 CFR 173 336 All cylinders must be the double connection type in order to facilitate cleaning.

b. Product may be shipped in cylinders of specification DOT 3A480 or 3AA480, or individual cylinders of specification DOT 106A500-X as specified in 49 CFR 173 336

c. Product may be shipped in cargo tanks as specified in DOT-E-3121

d Product may be shipped in tank cars of specification DOT 105A500-W as specified in 49 CFR 173 336

e The low-iron grade MON-1 and MON-3 product shall be shipped in stainless steel containers whenever practical

6.4 Definitions

a. Particulate. the undissolved solids in a given product retained on a 10 micron filter membrane

b. Single-phased liquid A single-phased liquid is devoid of any visible foreign liquid but may contain solid material as permitted within this specification

6.5 Highway Safety To promote safety in the transportation of propellants in interstate commerce by motor vehicle, each product contractor or shipper should assure (and provide if necessary) that each driver possesses DOT-P-5880 2 Guide 20 A complete manual of guides or the individual guide is available from the American Trucking Association, 1616 P Street, N W., Washington, D.C 20036

6.6 Precautionary Labels Precautionary labels are prepared in accordance with ANSI Z129.1 For those propellants which do not have specifically prescribed labels, the principles for the preparation of the labels are used There does not have to be exact agreement between labels from different sources as long as the intent of the manual is complied with

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6.7 Pollution Control. U S. Public Laws dictate increased effort to improve air, land, and water pollution control of toxic propellant vapors, leaks, spills, and disposal during all phases of manufacture, transfer, storage, and transportation operations The manufacturer/supplier is enjoined to approach the appropriate pollution control district to mutually resolve all problem areas, and to develop adequate control and disposal methods for situations which are likely to develop in any of the phases.

Custodians.
Army - MI
Navy - AS
Air Force - 68

Review Activities
Air Force - 12, 14, 19

Preparing Activity.
AF-68

Other Agency Interest:
NASA

(Project 9135-0123)

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	☐ MANUFACTURER
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5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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